

Latency Report

Pipeline: raw STT -> LLM -> TTS | Transport: non-streaming | Comparison: clean LLM mode vs speech directive mode

Result

Directive mode reduced the median tool-turn end-to-heard latency from 7.20s to 5.95s. The clearest cause is the model layer: tool-turn LLM time dropped from 2.85s median / 8.51s p95 to 1.84s median / 1.89s p95 because the post-tool response LLM pass is skipped.

Mode	Turns	Tool turns	All turns end-to-heard median / p95	Tool turns end-to-heard median / p95	Tool-turn LLM median / p95	Tool-turn TTS full median / p95
LLM mode	9	6	6.61s / 13.95s	7.20s / 13.95s	2.85s / 8.51s	2.14s / 2.93s
Directive mode	9	4	6.19s / 8.76s	5.95s / 6.78s	1.84s / 1.89s	1.99s / 3.07s

Findings

- End-of-turn detection did not explain the difference: both runs stayed around 0.68s median.
- STT was similar enough for this comparison: LLM 1.25s / 2.85s vs directive 1.37s / 4.35s median / p95.
- Local tool execution was effectively zero in both modes because tools call local in-process functions over sample data.
- TTS was not the main differentiator in the clean pass: tool-turn TTS full was 2.14s vs 1.99s median.
- Determinism improved in directive mode: instead of the model paraphrasing the tool outcome, the agent speaks the exact directive text returned by the backend.

Conclusion

The meaningful optimization is speech-directive mode for tool-backed turns. It keeps local tool latency unchanged, removes the second model generation step, and makes tool outcomes deterministic. The next optimization should target streaming STT/LLM/TTS so the browser can hear first audio before full synthesis completes.